Engine-Native AlignAtt for Decoder-Only LLM MT in an IWSLT 2026 Simultaneous Speech Translation System

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Abstract

We describe our IWSLT 2026 simultaneous speech translation system: a synchronous cascade with Qwen3 forced ASR on the speech side and Gemma-4 E4B-it with AlignAtt on the machine-translation side, evaluated on English-to-German, English-to-Chinese, and English-to-Italian in low-latency and high-latency regimes. The technical obstacle is that the MT model is a decoder-only LLM served through fused, graph-compiled inference, so the cross-attention signal required by standard AlignAtt is neither architectural nor directly observable at runtime. We recover a usable MT-side policy through a deterministic prompt layout, offline selection of translation-specific heads, selective *qk-fast* replay of the draft-to-source attention block, and an engine-native query/key capture path that preserves model outputs bit-identically. On the IWSLT 2026 dev set, the retained cascade reaches a low-latency operating point near two seconds of compute-unaware LongYAAL and trades 1.32–1.53 additional seconds for clear gains in BLEU and XCOMET-XL in the high-latency regime. Head selection remains an end-to-end quality/latency trade-off, and the deployed *qk-fast* path keeps the MT observer cheap enough to stay on the deployed runtime.

1 Introduction

This paper describes the retained IWSLT 2026 submission system for English-to-German, English-to-Chinese, and English-to-Italian. The stack is a synchronous Qwen3 forced ASR + Gemma-4 E4B-it cascade with two operating regimes: one low-latency and one high-latency. Earlier exploration of an all-Gemma "single substrate" variant remains in the appendix, but it is not part of the submitted system.

One deployment choice shapes the rest of the paper: both ASR and MT are served through vLLM so that speech and translation share a single compiled inference stack. On the MT side, that choice removes the standard AlignAtt observation point twice over. Gemma-4 is decoder-only, so there is no decoder cross-attention to read, and vLLM keeps self-attention inside fused, graph-compiled kernels, so the full attention matrix is not exposed to Python. The main technical contribution is therefore an engine-native AlignAtt realization for decoder-only MT: a deterministic prompt layout to mark the source span, offline head selection, selective replay of the draft-to-source block, and compile-safe query/key capture that preserves model outputs bit-identically.

Section 2 situates the problem, Section 3 describes the submitted system, Section 4 details the MT-side AlignAtt realization, Section 5 reports the results, and the appendix keeps the heavier derivations and exploratory analyses.

2 Background

AlignAtt was introduced for encoder–decoder simultaneous speech translation, where each decoder cross-attention row is normalized over source positions only (Papi et al., 2023). In that setting, source provenance is almost architectural: the online policy can inspect whether a drafted token depends on source frames that are already accessible or still too recent. That is the point of departure for our paper. We keep the same policy intuition, but we have to recover the alignment signal from a very different substrate.

On the MT side, the submitted system uses a decoder-only LLM rather than an encoder–decoder translation model. The source transcript, translation instruction, accepted target prefix, and drafted continuation all live inside one causal prompt, and the deployed runtime fuses attention inside graph-compiled kernels. Prior interpretability work nevertheless shows that multilingual LLMs contain a sparse set of translation-specific heads (Liu et al., 2026). This makes an

AlignAtt-style policy plausible, but only if the source span is made explicit in prompt space and the policy-visible attention rows can be reconstructed from tensors captured on the real inference path.

The evaluation protocol also matters for interpretation. We follow the Simulstream setting of long-form, unsegmented, revision-capable streaming translation (Gaido et al., 2025) and report LongYAAL in both compute-unaware and compute-aware variants (Polák et al., 2025). The submitted cascade itself is synchronous and single-GPU: each chunk triggers ASR first and MT second. This conservative regime cleanly separates policy behavior from scheduler overlap.

Figure 1 summarizes the conceptual shift. The policy intuition remains the same as in encoder–decoder AlignAtt, but the alignment signal must now be recovered from a prompt-structured self-attention substrate rather than read directly from cross-attention.

3 Submitted System Overview

ASR backend. The speech side of the submitted cascade uses Qwen3-ASR with the Qwen3 forced aligner (Qwen Team, 2025). For each chunk boundary, it returns an updated transcript prefix together with per-word end times. We retain this dedicated aligner because it is materially better than the discarded all-Gemma ASR path on boundary accuracy, word error rate, and runtime; the auxiliary comparison is reported in Appendix C.

MT backend. The translation side uses Gemma-4 E4B-it served through graph-compiled vLLM. At step k, Gemma receives the current transcript prefix, the already accepted target prefix, and a fixed translation instruction, then generates a short draft continuation. AlignAtt accepts the longest draft prefix whose alignment signal stays on the currently accessible side of the source frontier.

Streaming loop. The cascade is synchronous and chunk-based. Each chunk first updates the ASR transcript prefix, then launches one MT request. Source words are considered accessible only once their end time is older than a small hold-back margin. MT output is append-only: once a target token is accepted, it is reused as causal prefill for the next MT step and is never retracted.

Submitted operating points. We keep two official frozen submission regimes. The low-latency submission uses $\Delta_{\text{chunk}} = 850$ ms and the high-latency submission uses $\Delta_{\text{chunk}} = 1500$ ms. Both frozen bundles still include the historical MT rewind heuristic that was active at submission time, together with the same ASR backend, Gemma-4 MT backbone, and MT-side AlignAtt realization. Later no-rewind reruns are therefore treated in this paper only as auxiliary diagnostics rather than as submitted operating points.

4 Engine-Native AlignAtt for LLMs

This section focuses on the MT-side mechanism that makes Gemma-4 a usable AlignAtt backend inside the retained cascade.

4.1 Prompt Layout and Alignment Heads

Standard AlignAtt does not directly apply because a decoder-only LLM has no decoder cross-attention. We therefore recover a source-conditioned view by forcing a deterministic prompt layout:

$$\mathbf{p}^{(k)} = \left[\mathbf{p}^{\text{sys}} \| \mathbf{s}^{(k)} \| \mathbf{p}^{\text{instr}}, \mathbf{y}_{1:m_k} \| \widetilde{\mathbf{y}}^{(k)}\right], \quad (1)$$

where $\mathbf{s}^{(k)}$ is the live transcript prefix returned by ASR, $\mathbf{y}_{1:m_k}$ is the already accepted translation prefix, and $\widetilde{\mathbf{y}}^{(k)}$ is the draft proposed by Gemma-4 at the current step. The transcript prefix is therefore a contiguous prompt span with a known map $\phi^{(k)}$ from source-word indices to prompt positions.

Not all self-attention heads carry a useful translation signal (Liu et al., 2026). We score Gemma heads offline on held-out word-aligned MT examples under this exact prompt layout and retain the top $k = 8$ heads per language pair. This head set is fixed at inference time and is the only part of the policy that depends on offline calibration. Appendix A reports the calibration table and the full layer-head score maps.

4.2 Selective Reconstruction for the Policy

The policy never needs the full $n \times n$ self-attention matrix. It only needs draft rows against source columns. For selected heads (ℓ, h), draft positions t, and source positions s, we reconstruct

$$\widehat{A}_{t,s}^{(\ell,h)} = \frac{\exp\left(q_t^{(\ell,h)} \cdot k_{\phi^{(k)}(s)}^{(\ell,h)}\right)}{\sum_j \exp\left(q_t^{(\ell,h)} \cdot k_j^{(\ell,h)} + m_{t,j}\right)}, \quad (2)$$

where the denominator still ranges over the entire prompt. We call this replay path *qk-fast*: it is

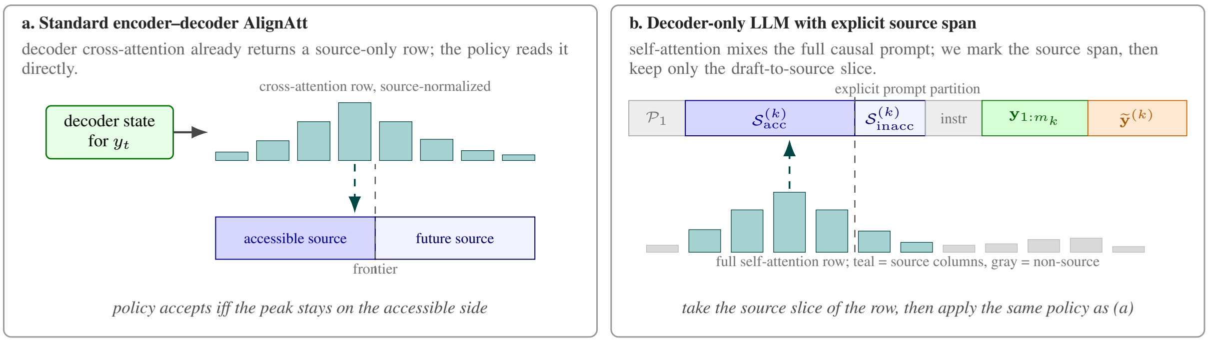


Figure 1: **How AlignAtt changes substrate between encoder–decoder and decoder-only models.** *(a)* In the original encoder–decoder setting, the decoder already exposes a source-only cross-attention row, so the policy can gate tokens directly against the accessible source frontier: if the peak of the row falls on the accessible side, the draft token is accepted. *(b)* In our decoder-only MT setting, source and target history share one causal prompt. We therefore mark the source span explicitly in prompt space, isolate translation-specific heads, and reconstruct only the draft-to-source slice needed by the policy; the decision rule is otherwise the same as in (a).

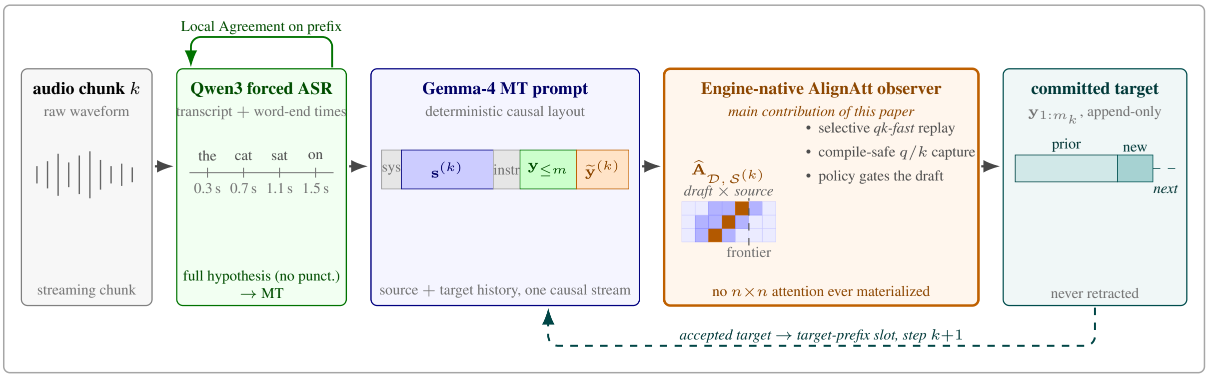


Figure 2: **Submitted cascade, one chunk-synchronous step.** Each chunk first updates the source prefix with Qwen3 forced ASR, then runs one Gemma-4 MT step. The MT prompt keeps the ASR transcript explicit inside the decoder-only causal layout, so the engine-native observer (*main contribution of this paper*) can capture the selected heads' queries and keys on the deployed vLLM path and reconstruct only the draft-to-source block $\widehat{\mathbf{A}}_{\mathcal{D},\mathcal{S}^{(k)}}$ used by the acceptance policy. Accepted target tokens are appended to $\mathbf{y}_{1:m_k}$ and become the target-prefix slot of the next step's prompt (dashed teal feedback).

exact up to floating-point reassociation relative to the fused forward and only reconstructs the policy-visible block.

From these per-head rows we build two source-side provenance features for each drafted token: accessible-source mass and inaccessible-source mass,

$$\pi_t^{\text{acc}} = \sum_{s \in \mathcal{S}^{(k)}_{\text{acc}}} \bar{\mathbf{r}}_t(s), \tag{3}$$

$$\pi_t^{\text{inacc}} = \sum_{s \in \mathcal{S}^{(k)}_{\text{inacc}}} \bar{\mathbf{r}}_t(s), \tag{4}$$

where $\bar{\mathbf{r}}_t$ is the head-averaged reconstructed row. We additionally normalize per-head rows online with prefix Welford statistics and apply a short median filter along the source axis before taking the source-side argmax. This stabilizes the alignment peak without changing the underlying replayed rows.

4.3 AlignAtt Acceptance Policy

The submitted MT policy is a first-failure scan over drafted tokens. Let $\hat{s}_t$ be the source-side argmax of the filtered row for draft token t, and let $N^{(k)}_{\text{acc}}$ be the number of source words whose ASR end time is already on the accessible side of the frontier. The scan stops when one of three conditions first fires:

$$\text{SOURCE-FRONTIER}: \quad \hat{s}_t \geq N^{(k)}_{\text{acc}} + b, \tag{5}$$

$$\text{ARGMAX-MASS-WEAK}: \quad \bar{\mathbf{r}}_t(\hat{s}_t) < \tau_{\text{argmax}}, \tag{6}$$

$$\text{PROVENANCE-WEAK}: \quad \pi_t^{\text{acc}} < \tau_{\text{src}}. \tag{7}$$

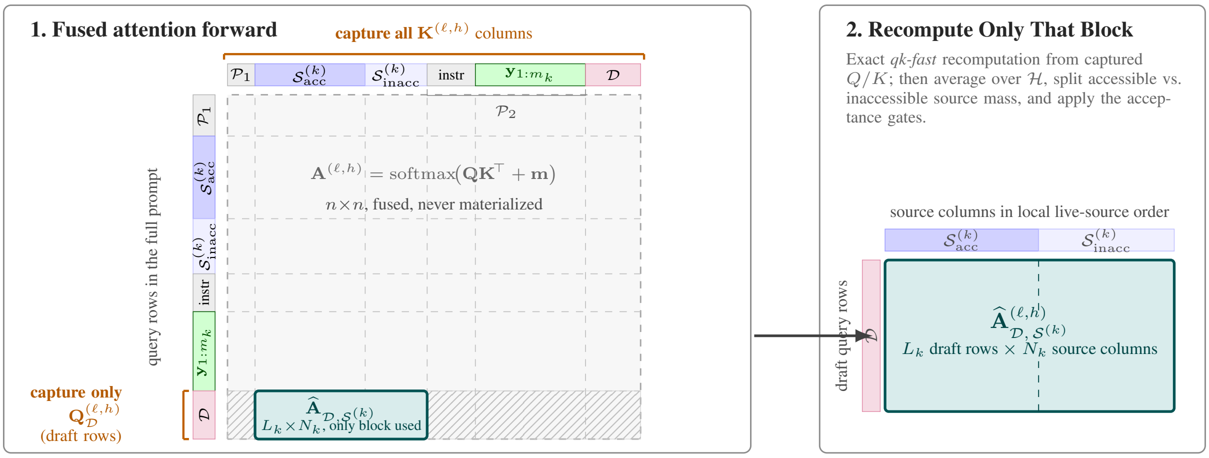


Figure 3: **Selective reconstruction with engine-native capture.** The $n \times n$ attention matrix $\mathbf{A}^{(\ell,h)}$ is executed entirely inside the fused attention kernel, so the full $n \times n$ matrix is never materialized. For each AlignAtt head $(\ell, h) \in \mathcal{H}$, the observer copies $\mathbf{K}^{(\ell,h)}$ for all positions and $\mathbf{Q}^{(\ell,h)}$ only for the draft rows $\mathcal{D}$ into fixed-shape buffers. The green band $\mathbf{y}_{1:m_k}$ inside $\mathcal{P}_2$ marks the committed target prefix that grows across chunks; hatched cells are never reconstructed. After the forward, Eq. (2) recomputes exactly only the draft-to-source block $\widehat{\mathbf{A}}_{\mathcal{D},\mathcal{S}^{(k)}}$, which is the sole object consumed by the source-side provenance scores of Eqs. (3)–(4) and by the acceptance policy of this section.

The first condition prevents dependence on still-inaccessible source words, the second rejects weak alignment peaks, and the third rejects tokens with too little total accessible-source support. The accepted output is the longest draft prefix preceding the first failure, rounded back to the nearest stability-unit boundary so partial subword fragments are never emitted. We intentionally do not impose an anti-rewind rule on $\hat{s}_t$, because that bias consistently hurt valid reorderings in EN→ZH development.

The frozen submitted bundles additionally wrap this frontier rule in a small historical rewind heuristic on the source-side argmax trajectory. We later found that removing this guard helps language pairs with stronger reorderings, especially EN→ZH, so rewind should be read here as a property of the frozen submission artifacts rather than of the underlying AlignAtt formulation. Figure 4 shows the gate on a live MT draft: black stars mark drafted tokens whose source argmax stays on the accessible side of the frontier, and the first red star marks the position at which the SOURCE-FRONTIER condition fires.

4.4 Engine-Native Capture under Compiled Inference

Equation (2) is only useful if the deployed runtime exposes the exact queries and keys consumed by attention. Under fused, graph-compiled vLLM, those tensors never appear as ordinary Python-visible objects. We therefore install a per-layer observer before compilation, copy prompt keys and draft-row queries into fixed-shape buffers, and route capture through a custom-op-style call that survives graph lowering. A zero-valued sentinel dependency keeps the observer in the transitive fan-in of the graph output:

$$\mathbf{o}'_\ell = \mathbf{o}_\ell + \Phi\big(\ell, \mathrm{pos}, \mathbf{Q}^{(\ell)}, \mathbf{K}^{(\ell)}\big), \qquad \Phi \equiv \mathbf{0}. \quad (8)$$

This leaves model outputs unchanged while making the policy-visible rows recoverable on the actual submission path. Appendix B gives the full buffer layout, replay details, parity measurements, and qualitative reconstruction examples.

5 Results

5.1 Experimental Setup

All experiments run on a single NVIDIA A40. The MT backend uses Gemma-4 E4B-it in bfloat16 through graph-compiled vLLM; the speech side uses Qwen3-ASR with forced alignment. Evaluation uses the official 21-clip IWSLT 2026 MCIF dev set ($\sim$ 2.1 hours total, long academic talks) resegmented by OmniSTEval v0.1.7. We report BLEU and chrF against the dev references, XCOMET-XL for translation adequacy (Guerreiro et al., 2024), and compute-unaware / compute-aware LongYAAL for latency (Polák et al., 2025). The streaming loop follows the

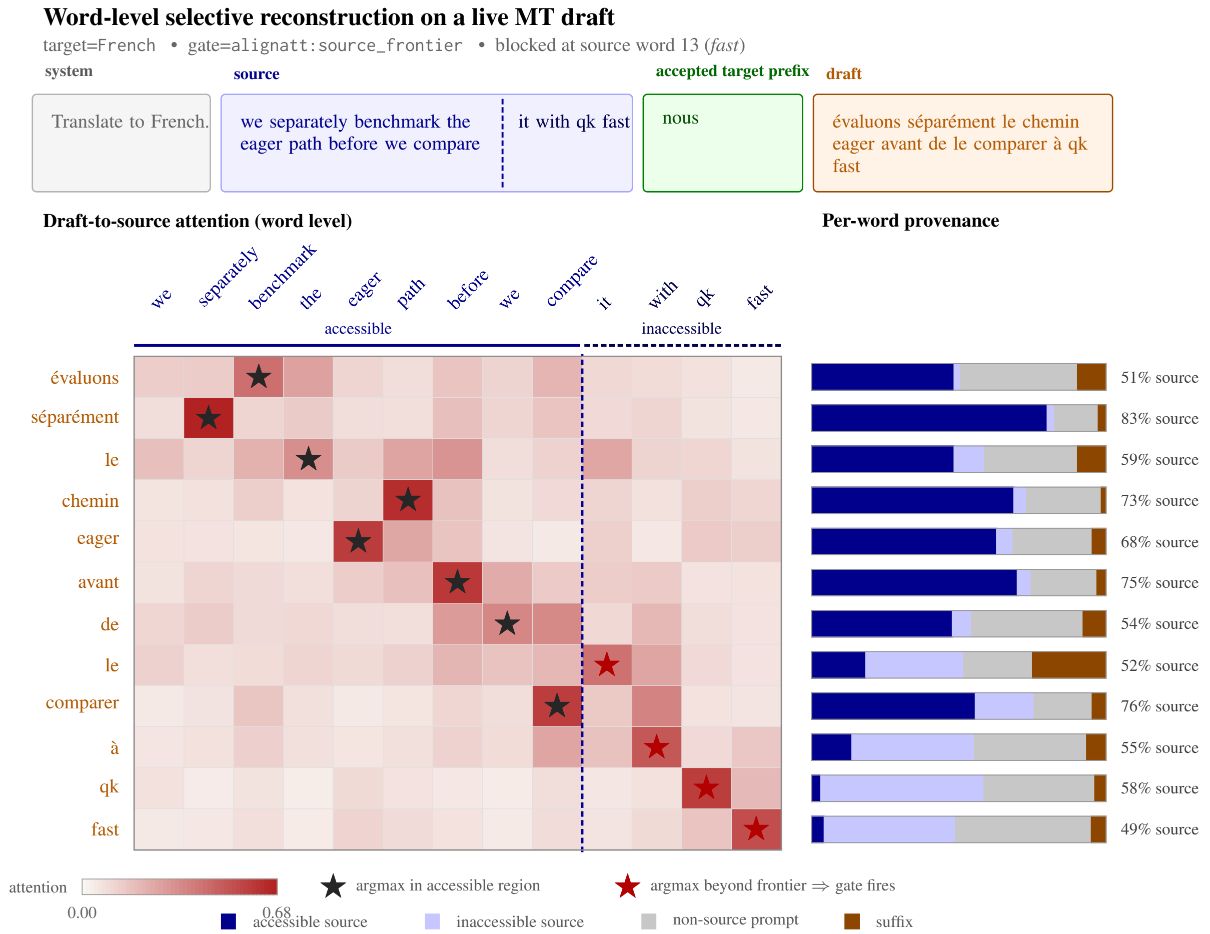


Figure 4: **Word-level selective reconstruction on a live MT draft.** *Top ribbon:* the decoder-only prompt is split into the system instruction, the live source (with the accessibility frontier shown in-line between the committed accessible prefix and the still-inaccessible tail), the already accepted target prefix reused from earlier streaming steps, and the draft currently under inference. *Left panel:* reconstructed draft-to-source attention from the selected AlignAtt heads, aggregated to the word level; rows are drafted words, columns are source words split by the dashed frontier. A star marks the per-row argmax (black when it lands in the accessible region, red when it lands beyond the frontier, which is exactly what triggers the SOURCE-FRONTIER gate). *Right panel:* for each drafted word the reconstructed attention mass is partitioned into the four provenance classes; the numeric annotation is the total source share (accessible + inaccessible). The green band is previously committed target text reused as causal context; it is not itself the object of the gate. The first black stars in the orange draft show that newly proposed words may still align to multiple source words inside the current accessible span, and only the first red star beyond the frontier triggers rejection.

Simulstream setting of unsegmented long-form audio with revision-capable evaluation (Gaido et al., 2025), but our submitted cascade itself is synchronous: ASR and MT are serialized on one GPU so that policy effects are not confounded with scheduler overlap.

For the auxiliary runtime study in Section 5.4, we compare three MT seams on a fixed suite of 16 text-only prompts: a minimal Transformers eager reference, a Transformers *qk-fast* replay reference, and the deployed vLLM *qk-fast* path. Each seam uses greedy decoding with the same head set, one warmup pass, and repeated hot runs in isolated subprocesses.

5.2 Submitted System Results

Table 1 reports the frozen submitted bundles. In the low-latency regime, all three language pairs still cluster tightly around 2.0 s CU-LongYAAL: 2.00 s for EN→DE, 1.95 s for EN→ZH, and 1.98 s for EN→IT. Moving to the high-latency regime costs 1.32–1.53 additional seconds, but adds 3.85–4.36 BLEU and 0.027–0.036 XCOMET-XL. The main end-to-end result is therefore simple: the submitted cascade delivers a stable low-latency operating point, and the MT-side AlignAtt realization lets the same decoder-only Gemma backend move up the quality/latency frontier when more latency is allowed.

5.3 Head Filtering Matters End-to-End

Table 2 is an auxiliary EN→DE rerun included only to isolate the MT head-set effect under the maintained *qk-fast* runtime, without the historical MT rewind heuristic used by the frozen submission bundles. End-to-end quality is nearly equivalent: replaying all 336 heads improves XCOMET-XL by only +0.0059 relative to the retained top-8 setup, while increasing CU-LongYAAL by +100.3 ms and CA-LongYAAL by +179.5 ms. This is the expected *qk-fast* trade-off because replay cost grows with the number of reconstructed heads. We therefore read the auxiliary comparison as evidence that a small retained head set captures most of the useful MT alignment signal at much lower runtime cost. Appendix A gives the held-out calibration numbers behind the retained head sets.

5.4 Runtime Cost of the MT Observer

The MT observer must be cheap enough to justify deployment. Figure 5 shows that the engine-native vLLM replay path is much cheaper than the minimal eager seam we use as an inspection reference: median cost drops from 63.7 to 25.4 ms/token, while the Transformers-side *qk-fast* replay remains close to the eager baseline at 59.0 ms/token. This confirms that the MT contribution is not merely inspectable in principle; it is practical inside the compiled runtime we actually deploy.

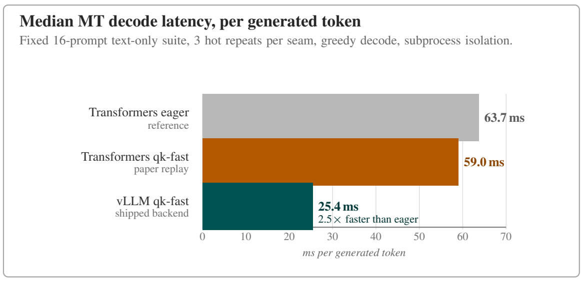


Figure 5: **Inference-time comparison of MT capture seams.** Median latency per generated token on a fixed 16-prompt text-only suite, for a minimal Transformers eager reference, a Transformers *qk-fast* reference that reconstructs source rows from captured layer inputs, and the engine-native vLLM *qk-fast* path used by the shipped backend. Each seam uses one unmeasured warmup pass followed by three hot repeats, greedy decode, a fixed AlignAtt head set, and subprocess isolation; per-prompt results and divergences are stored in the generated JSON artifact.

5.5 Coordination Regimes and Compute-Aware Latency

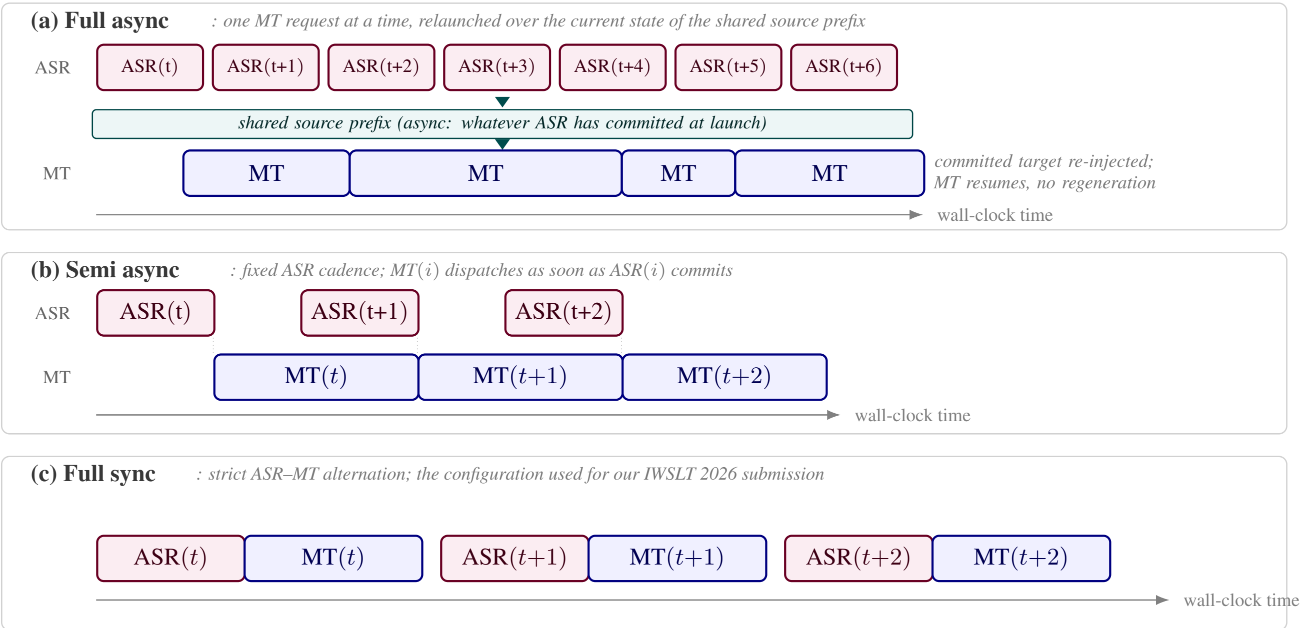
Our cascade loads the ASR and MT models as two separate blocking `vllm.LLM` instances on a single GPU, which serializes every audio chunk into an ASR decode followed by an MT decode (Fig. 6 (c)). This conservative regime keeps policy effects separate from scheduler overlap. Alternative pipelined or fully asynchronous regimes would mainly reduce compute-aware LongYAAL at fixed compute-unaware latency, but would require a non-blocking scheduler and a more careful treatment of stale MT requests. We leave that scheduler axis to future work.

6 Conclusion

The retained IWSLT 2026 system is a Qwen3 forced ASR + Gemma-4 MT cascade, and its key technical ingredient is an engine-native realization of AlignAtt for decoder-only MT under compiled inference. Deterministic prompt layout, offline head selection, selective *qk-fast* replay, and

Pair	Regime	BLEU	chrF	XCOMET-XL	LongYAAL (s)	
					CU	CA
en→de	low	28.76	62.1	0.875	2.00	1.63
	high-latency	32.63	64.2	0.902	3.53	3.14
en→zh	low	36.01	35.0	0.743	1.95	1.77
	high-latency	39.86	37.8	0.778	3.27	3.09
en→it	low	40.10	68.0	0.805	1.98	1.62
	high-latency	44.46	70.1	0.841	3.48	3.10

Table 1: **Submitted cascade on the IWSLT 2026 dev set** (21 clips, OmniSTEval long-form resegmentation). Low-latency rows use the frozen submitted setting with $\Delta_{\text{chunk}} = 850$ ms and the historical MT rewind policy. High-latency rows use the frozen submitted setting with $\Delta_{\text{chunk}} = 1500$ ms and the same rewind-equipped MT runtime.



Regimes (a) and (b) require a non-blocking scheduler (e.g. vLLM `AsyncLLMEngine`) with request abort on stale MT decodes; our current implementation is (c) with two blocking `LLM` instances sharing a single GPU. The gaps between ASR chunks in (a) and (b) reflect an ASR real-time factor below 1. This design axis mainly affects *compute-aware* latency; our central submitted comparisons remain on the serialized regime in (c).

Figure 6: Three coordination regimes for ASR and MT sharing one GPU. The submitted system uses the strict ASR–MT alternation in (c).

Setting	XCOMET-XL ↑	CU (s) ↓	CA (s) ↓
Top-8 heads	0.879	**1.96**	**1.65**
All 336 heads	**0.885**	2.06	1.83

Table 2: **Auxiliary EN→DE MT head-set comparison at $\Delta_{\text{chunk}} = 1100$ ms.** Both rows use the same maintained runtime without the historical MT rewind heuristic; only the MT head set changes. End-to-end quality is nearly equivalent, but all-head replay increases CU and especially CA because the runtime must reconstruct more attention at each MT step.

compile-safe query/key capture make the MT policy usable without leaving the deployed runtime. On the dev set, this yields a low-latency operating point near 2 s CU-LongYAAL and a high-latency regime that is roughly 1.5 s slower but clearly stronger in BLEU and XCOMET-XL. The main limitations are that evaluation is single-GPU and dev-set only, and that the contribution is on the MT observer rather than on speech modeling itself.

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A Additional Decoder-Only AlignAtt Details

From encoder–decoder AlignAtt to decoder-only MT. In its original formulation, AlignAtt exploits decoder cross-attention in an encoder–decoder speech translation model (Papi et al., 2023): each decoder row is normalized over source positions only, so source provenance is built into the architecture. In a decoder-only LLM there is no cross-attention. A single causal token stream mixes instructions, source, accepted target history, and the current draft, and each row of self-attention is normalized over all prompt positions. Relative to the encoder–decoder setting, the attention row no longer distinguishes source from non-source by construction, and no head is architecturally reserved for source alignment.

Prompt-induced source span. We therefore recover a source-target distinction through prompt layout rather than architecture. At MT step k, the prompt is assembled as

$$\mathbf{p}^{(k)} = \left[\, \mathbf{p}^{\text{sys}} \| \mathbf{s}^{(k)} \| \mathbf{p}^{\text{instr}}, \mathbf{y}_{1:m_k} \| \widetilde{\mathbf{y}}^{(k)} \,\right], \tag{9}$$

where $\mathbf{s}^{(k)}$ is the live transcript prefix, $\mathbf{y}_{1:m_k}$ the accepted target prefix, and $\widetilde{\mathbf{y}}^{(k)}$ the drafted continuation. Let $\phi^{(k)} : \mathcal{S}^{(k)} \hookrightarrow \{1, \ldots, n\}$ map local source indices to absolute prompt positions.

Virtual source-restricted attention rows. The object relevant to alignment is then the source slice of self-attention. For draft position t and source index s, we read

$$\mathbf{A}_{\text{src}}^{(\ell,h)}(t, s) := \mathbf{A}_{t,\,\phi^{(k)}(s)}^{(\ell,h)}, \tag{10}$$

at layer ℓ and head h. Unlike encoder–decoder cross-attention, $\sum_{s \in \mathcal{S}^{(k)}} \mathbf{A}_{\text{src}}^{(\ell,h)}(t, s) < 1$ in general because the row is normalized over the entire prompt. Residual mass on non-source prompt positions and on the draft is therefore structural, not an implementation artifact.

What stays the same and what changes. The online decision rule still follows the original AlignAtt spirit: accept the longest draft prefix whose source-side alignment signal remains inside the accessible frontier. What changes is the substrate on which that signal is observed. In the decoder-only case, we carve source-restricted rows out of the full-prompt self-attention tensor and keep track of the residual mass on the accepted target prefix, the rest of the prompt, and the speculative suffix.

Alignment heads. Not all self-attention heads carry translation alignment signal. We assume the existence of a small head subset $\mathcal{H} \subset \{1, \ldots, L\} \times \{1, \ldots, H\}$, $|\mathcal{H}| \ll LH$, such that the head-averaged source row approximates gold source-to-target alignments on a held-out calibration set $\mathcal{G} = \{(\mathbf{s}^{(i)}, \mathbf{y}^{(i)}, \mathbf{a}^{(i)})\}_i$, where $\mathbf{a}^{(i)}$ is a forced word-level alignment. This sparsity assumption is consistent with recent evidence that multilingual LLMs contain a small set of translation-specific alignment heads (Liu et al., 2026).

Offline head selection. For one head (ℓ, h), define the argmax alignment

$$\hat{a}_t^{(\ell,h)}(\mathbf{s}, \mathbf{y}) = \arg\max_s \mathbf{A}_{t,\phi(s)}^{(\ell,h)}, \tag{11}$$

and its token-level alignment accuracy

$$\begin{aligned} \mathrm{Acc}_{\text{align}}(\ell, h) = \frac{1}{|\mathcal{G}|} \sum_i \frac{1}{|\mathbf{y}^{(i)}|} \\ \cdot \sum_{t=1}^{|\mathbf{y}^{(i)}|} \mathbf{1}\big[\hat{a}_t^{(\ell,h)} = \mathbf{a}_t^{(i)}\big]. \end{aligned} \tag{12}$$

We then select $\mathcal{H} = \text{top-}k_{(\ell,h)} \mathrm{Acc}_{\text{align}}(\ell, h)$ with $k = 8$.

Pair	Top-8 TS	All-336 TS	Gain	Aligned tokens
EN→DE	90.40	68.49	**+21.91**	11,209
EN→ZH	93.48	65.79	**+27.70**	7,582
EN→IT	91.90	67.42	**+24.49**	12,056

Table 3: **MT head-set filtering ablation on held-out word-aligned dev examples.** To make the all-head baseline maximally charitable, we average the candidate head set first and then restrict the argmax to the source-prompt token zone before scoring against gold aligned source tokens. Scores are reported in points (100 × TS). Even under this source-only argmax, the per-direction retained top-8 heads still preserve a markedly stronger alignment signal than uniform all-head averaging.

Before testing these head sets end-to-end in Section 5.3, we first score them on held-out word-aligned MT examples under the same prompt layout. Table 3 shows that the retained top-8 heads preserve a markedly stronger alignment signal than uniform all-head averaging even when the all-head baseline is made maximally charitable by restricting its argmax to the source-prompt zone before scoring.

One abstraction, two roles. The same abstraction can in principle be used for ASR and MT. In the retained submission we deploy it only on the MT side, but the role-agnostic formulation still explains why the same alignment toolkit was useful during development: only the prompt layout and the selected head set change between the two roles.

B Observer Implementation, Proof Sketches, and Qualitative Diagnostics

Method overview. Section 4 gives the compact main-paper version. This appendix keeps the fuller derivations, proof sketches, and runtime details. The method has two coupled parts. First, we reconstruct exactly only the draft-to-source attention block consumed by the policy. Second, we make the required query and key tensors observable inside a fused, graph-compiled runtime where the full attention matrix never appears in memory.

***qk-fast* selective reconstruction.** Let $\mathcal{H}$ be the selected MT head set. The acceptance policy never consumes the full $n \times n$ matrix; it only needs draft rows against source columns. We therefore reconstruct only that block:

$$\widehat{\mathbf{A}}_{t,s}^{(\ell,h)} = \frac{\exp\big(\mathbf{Q}_t^{(\ell,h)} \cdot \mathbf{K}_{\phi^{(k)}(s)}^{(\ell,h)}\big)}{\sum_{j=1}^{n} \exp\big(\mathbf{Q}_t^{(\ell,h)} \cdot \mathbf{K}_j^{(\ell,h)} + m_{t,j}\big)}, \tag{13}$$

for draft positions t, source positions s, and heads $(\ell, h) \in \mathcal{H}$. The denominator still ranges over the entire prompt, so the runtime must expose keys for all prompt positions even though the reconstructed object is indexed only by source positions.

Why this simplification is safe. Two qualitative facts matter more than the omitted derivations. First, the source-side gate is frontier-monotone: when ASR reveals more source words, mass can only move from the inaccessible bucket to the accessible bucket, and the allowed frontier only moves right. Second, the replay noise we observe is small relative to typical post-filter argmax margins, which is why the eager and engine-native paths make the same acceptance decisions on our parity suite. The cost story is similarly simple: full attention would scale with the whole LHn^2 tensor, whereas replay only scales with the selected heads, the draft length, and the live prompt length. With $|\mathcal{H}|$=8 and short drafts, that selective path stays cheap enough to run online.

Why runtime observability is part of the method. Equation (13) is only useful if the runtime exposes the selected query and key tensors. Under fused attention kernels and `cudagraph=full`, the Q/K tensors never leave the fused kernel, Python-side recorders cannot hook into the compiled graph, and unused outputs are eliminated during lowering. The observer is therefore not ancillary tooling; it is the mechanism that makes selective reconstruction available in the actual inference stack.

Compile-safe capture and replay. The observer is installed before compilation, one slot per layer, so the compiled graph never encounters missing attributes. During the fused forward, it records only fixed-shape prompt-key and draft query/key tensors for the selected head subset; this is why the capture path stays compile-safe. The custom op is numerically inert, but its zero-valued return is added back into the attention output so the compiler cannot dead-code-eliminate it. After the decode step, the stored buffers are sufficient to rebuild prompt-side logits, suffix-side logits, and

	Qwen3 forced	Gemma AlignAtt
WER ↓	0.113	0.269
CER ↓	0.046	0.122
Boundary F1 $\pm$3w ↑	0.878	0.042
Mean |lag| (s) ↓	0.145	0.403
Load time (s)	63.7	123.0
RTF (wall-clock) ↓	0.306	0.628

Table 4: **ASR alignment comparison on `ccpXHNfaoy` (360 s).** Qwen3 forced remains clearly stronger than the discarded Gemma AlignAtt ASR path on both boundary quality and runtime.

finally the same source-restricted rows that the policy consumes.

Qualitative reconstruction example. Figure 4 instantiates the full reconstruction stack on a text-only MT probe: the exact prompt partition seen by the decoder is shown, reconstructed token-level rows are aggregated to the word level for readability, and the four-way provenance accounting is kept visible for every drafted word.

Numerical parity. Let $\mathbf{A}^{(\mathrm{TF})}$ denote the reference attention tensor produced by an uncompiled Transformers forward on the identical prompt. On a curated parity set, we observe

$$\begin{aligned} \|\Delta\mathbf{A}\|_\infty \le 1.2 \times 10^{-2}, \\ \|\Delta\mathbf{A}\|_1/(nL_k) \le 4 \times 10^{-4}, \end{aligned} \tag{14}$$

with bit-identical acceptance decisions between the eager and compiled paths. The observer is therefore not just fast enough to keep, but faithful enough to support controlled MT ablations.

C Additional ASR Analyses

This appendix collects the all-Gemma ASR analyses that informed the final system choice but are no longer central to the retained submission narrative.

Gemma AlignAtt ASR is feasible but not competitive. We verified that the same AlignAtt toolkit can be made to run on the ASR side, but the resulting Gemma-ASR path is not competitive enough for submission. Table 4 compares it with the dedicated Qwen3 forced aligner on a diagnostic clip: Qwen3 forced is clearly better in error rate, boundary lag, and wall-clock RTF. We therefore keep Gemma AlignAtt ASR only as an exploratory result and retain Qwen3 forced ASR for the submitted system.